

Maths Algorithms part 3

DAY 82
30/09/2025

Divisibility & large Numbers

check if a large number is divisible by 3 or not

Given a number, the task is that we divide number by 3. The input number may be large and it may not be possible to store even if we use long long int

```
Input : n = 769452
Output : Yes
Input : n = 123456758933312
Output : No
Input : n = 3635883959606670431112222
Output : Yes
```

Since input number may be very large, we cannot use $n \% 3$ to check if a number is divisible by 3 or not, especially in languages like C/C++. The idea is based on following fact.

A Number is divisible by 3 if sum of its digits is divisible by 3.

For Example $n = 1332$
Sum of digits = $1 + 3 + 3 + 2$
= 9

Since sum is divisible by 3
answer is yes

Let us consider 1332, we can write it as $1332 = 1 \times 1000 + 3 \times 100 + 3 \times 10 + 2$. The proof is based on below observation: Remainder of 10^i divided by 3 is 1. So powers of 10 only result in value 1. Remainder of " $1 \times 1000 + 3 \times 100 + 3 \times 10 + 2$ " divided by 3 can be written as : $1 \times 1 + 3 \times 1 + 3 \times 1 + 2 = 9$. The above expression is basically sum of all digits. Since 9 is divisible by 3, answer is yes.

```
// Function to find that number
// divisible by 3 or not
static boolean check(String str)
{
    // Compute sum of digits
    int n = str.length();
    int digitSum = 0;
    for (int i=0; i<n; i++)
        digitSum += (str.charAt(i)-'0');

    // Check if sum of digits is
    // divisible by 3.
    return (digitSum % 3 == 0);
}

// main function
public static void main (String[] args)
{
    String str = "1332";
    if(check(str))
        System.out.println("Yes");
    else
        System.out.println("No");
}
```

check if a large number is divisible by 4 or not

given a number, the task is to check if a number is divisible by 4 or not. The input number may be large and it may not be possible to store even if we use long long int.

Input : $n = 1124$

Output : Yes

Input : $n = 1234567589333862$

Output : No

Input : $n = 363588395960667043875487$

Output : No

Since input number may be very large, we cannot use $n \% 4$ to check if a number is divisible by 4 or not,

A number is divisible by 4 if number formed by last two digits of it is divisible by 4. For example, let us consider 76952. Number formed by last two digits = 52. Since 52 is divisible by 4, answer is yes.

How does this work? Let us consider 76952, we can write it as $76952 = 7 \times 10000 + 6 \times 1000 + 9 \times 100 + 5 \times 10 + 2$

The proof is based on below observation:

Remainder of 10^i divided by 4 is 0 if i greater than or equal to two. Note that 100, 1000, .. etc lead to remainder 0 when divided by 4. So remainder of " $7 \times 10000 + 6 \times 1000 + 9 \times 100 + 5 \times 10 + 2$ " divided by 4 is equivalent to remainder of following:

$$0 + 0 + 0 + 5 \times 10 + 2 = 52$$

Therefore we can say that the whole number is divisible by 4 if 52 is divisible by 4.

```
// Function to find that number
// is divisible by 4 or not
static boolean check(String str)
{
    int n = str.length();

    // Empty string
    if (n == 0)
        return false;

    // If there is single digit
    if (n == 1)
        return ((str.charAt(0) - '0') % 4 == 0);

    // If number formed by last two digits is
    // divisible by 4.
    int last = str.charAt(n - 1) - '0';
    int second_last = str.charAt(n - 2) - '0';
    return ((second_last * 10 + last) % 4 == 0);
}
```

// Driver code

```
public static void main(String[] args)
{
    String str = "76952";

    // Function call
    if (check(str))
        System.out.println("Yes");
    else
        System.out.println("No");
}
```

check if the large Number is divisible By 6 or not

Given a number, the task is to check if a number is divisible by 6 or not. The input number may be large and it may not be possible to store even if we use long long int

Input : n = 2112

Output: Yes

Input : n = 1124

Output : No

Input : n = 363588395960667043875487

Output : No

Since input number may be very large, we cannot use $n \% 6$ to check if a number is divisible by 6 or not. Especially in languages like C/C++

A Number is divisible by 6 if it is divisible by 2 and 3.

a) a Number is divisible By 2 if its last digit is divisible by 2

b) A Number is divisible By 3 if Sum of digits is divisible by 3

```
// Function to find that number divisible by 6 or not
```

```
static boolean check(String str)
```

```
{
```

```
    int n = str.length();
```

```
    // Return false if number is not divisible by 2.
```

```
    if ((str.charAt(n-1) - '0') % 2 != 0)
```

```
        return false;
```

```
    // If we reach here, number is divisible by 2.
```

```
    // Now check for 3.
```

```
    // Compute sum of digits
```

```
    int digitSum = 0;
```

```
    for (int i=0; i<n; i++)
```

```
        digitSum += (str.charAt(i) - '0');
```

```
    // Check if sum of digits is divisible by 3
```

```
    return (digitSum % 3 == 0);
```

```
}
```

```
// main function
```

```
public static void main (String[] args)
```

```
{
```

```
    String str = "1332";
```

```
    if(check(str))
```

```
        System.out.println("Yes");
```

```
    else
```

```
        System.out.println("No");
```

```
}
```